Science of Sintering, **58** (2026) 409-417

https://doi.org/10.2298/SOS250428030N

UDK: 519.872; 622.765.5

Cross Sections Set and Transport Coefficients for Ar^+ Ions in Ar Gas

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Abstract:

In this work we present a complete cross sections dataset and transport coefficients of Ar^+ ions in Ar gas. These data are needed for modeling in numerous applications of technological importance. A Monte Carlo simulation method is applied to accurately calculate transport parameters in hydrodynamic regime. We discuss new data for Ar^+ ions in Ar gas where the drift velocity, characteristic energy, flux and bulk values of reduced mobility, rate coefficients, diffusion coefficients and other transport parameters are given as a function of low, moderate and high reduced electric fields E/N (E-electric field strength, N-gas number density). Furthermore, it is shown that the ion energy probability function follows the Maxwellian distribution at low electric fields, whereas at high fields, it deviates significantly.

Keywords*: Monte Carlo Simulations; Ar^+ions, Ar gas.*

1. Introduction

Cold plasmas are frequently used in new technologies where they open up the possibilities of non-intrusive production or modification of various substances [1]. These plasmas have a high electron temperature and low gas temperature so non-equilibrium behavior of a large number of species becomes important [2]. Current computer resources allow studies of complex global models [3] which describe the behavior of such plasmas by taking into account a very large number of particles. The knowledge of ion-neutral reactions is generally available [4] although the effects of reactions on transport parameters of particular ions are much less studied due to nondetectability of rapidly vanishing ionic fluxes. This especially holds for ions whose transport is affected by fast reactions [5, 6].

Transport of Ar^+ plays a significant role in various etching and deposition processes [7], in dark matter detection [8] and in many other scientific and technological applications. In this letter**,** the objective will be to study transport of Ar^+ in Ar gas. Since quantitative information about ion-molecule reactions is essential for understanding the production and transport of ions in electrical discharges**,** it is necessary to have the cross section set.

Since no direct information is found in the literature on how the mobility of inert ions**,** such as Ar^+ ions**,** behaves in Ar we also calculated transport parameters by using the Monte Carlo simulation technique and a complete cross section. Ions are tracked in infinite space, receive energy from a static electric field, and dissipate it in collisions with neutral gas particles. Collisions are instantaneous and isotropic, and the collision trajectory is determined by the null-collision method [9]. The model can be used to optimize the conditions in which the lead concentrate sintering procedure is carried out. The obtained simulation results were compared with the experimental results and they coincide to a considerable extent [10].

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Theory of Iterated Function Systems and Voronoi tessellation have been used for modeling PVDF porous structures [11] and computer simulations have shown a relation between lattice parameter and inversion parameter [12]. An initial set of cross sections for this system with more details is given in our paper [13].

2. Materials and Experimental Procedures

2.1 Calculation of the cross section sets

Complete cross section sets for ion transport are scarce in spite of a broad range of specific methods relevant for quantification of particular cross sections. The main problem in heavy particle scattering, easily and precisely selecting the state of the projectile and target before the collision, is still very complicated for a range of conditions, so databases for ion scattering [14, 4], are still devoid of such data. Phelps established the first worldwide accessible database with cross section sets [15] tested for each particular case either for swarm conditions of spatially resolved measurements of emission or ion mobility values. Another range of cross section sets was established by measurements of ionic transport coefficients [4] and this work is ongoing. In all cases, only the most important cross sections may be established from the transport data. In the following section we will establish a complete cross section set for Ar^+ scattering on Ar from 0.1 meV to 1000 eV which will be used to calculate transport properties. Generally, one may distinguish three characteristic energy ranges: the low-energy regime where polarization scattering is dominant, the medium-energy regime where the polarization scattering is gradually replaced by the hard sphere repulsion, and the high-energy approximation regime.

In Fig. 1 we present complete cross sections for collisions of Ar^+ with Ar versus laboratory energy of Ar^+ of Ar. The symbols and collisional processes are: isotropic elastic cross section σ_i, backward elastic σ_b, UV resonance line excitation σ_{uv}, 488 nm line excitation, 811 nm line excitation, ionization σj and Langevin cross section σ_L.

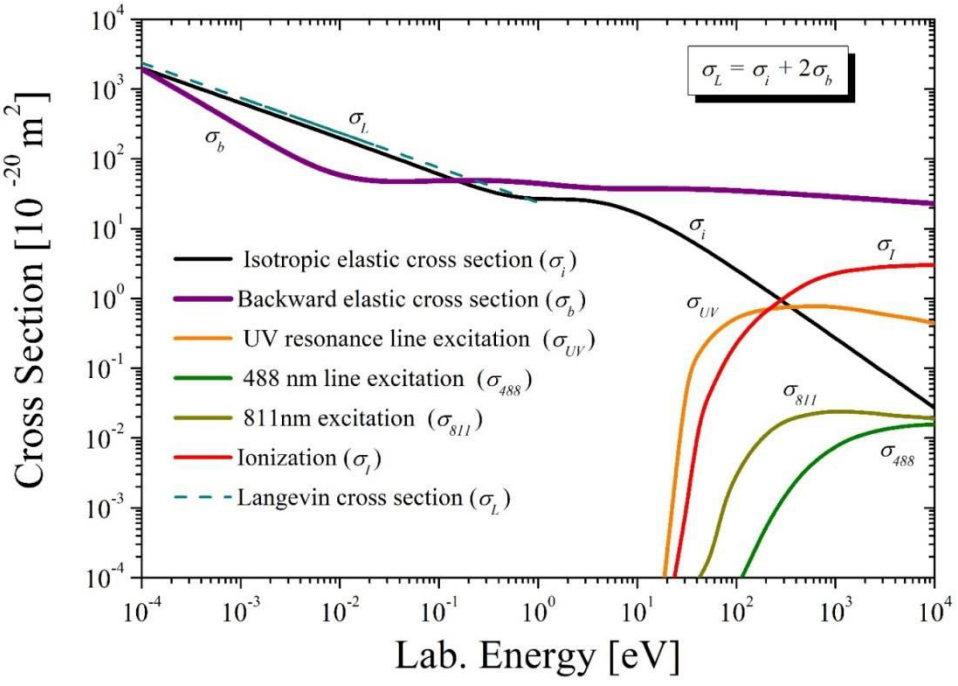


Fig. 1. Cross section set for Ar^+ + Ar.

3. Results and Discussion

The cross section sets for ion transport are rare due to a broad range of specific methods relevant for quantification of particular cross-sections. We have used a Monte Carlo code that properly takes into account thermal collisions [16]. The code has passed all the relevant benchmarks [17] and has been tested in our work on several types of charged particles [17, 18].

The results of Monte Carlo simulations are presented in Figs. 2–9. Fig. 2 shows the rate coefficients for collisions of Ar^+ ions with Ar gas at a temperature of T = 300 K, as a function of the reduced electric field E/N (E – electric field, N – number density of the gas). The dependence of transport properties on E/N arises directly from the structure of the Boltzmann equation and its solution, which ensures theoretical consistency and enables experimental comparison across a broad range of conditions.

The rate coefficient is defined as $k_i = dn_i / ndt$, where dn_i denotes the number of collisions of type i occurring within the time interval dt, and n represents the total number of ions in the simulations. Among the key results of the simulations are the rate coefficients for various processes, including the isotropic elastic cross section σ_i, backward elastic cross section σ_b, cross sections for UV resonance line excitations σ_{uv}, excitations of the 488 nm (σ_{488}) and 811 nm lines (σ_{811}), ionization σ_I, and the Langevin cross section σ_L. These cross sections serve as essential input parameters for fluid equations governing ion transport in the gas. The analysis of collision dynamics reveals that the backward elastic scattering process, which results in the formation of fast Ar atoms and slow Ar^+ ions, increases significantly with E/N, becoming the dominant process when E/N > 200 Td. Additionally, the rate coefficient for ionization becomes appreciable only at E/N > 2000 Td, which explains the absence of observable differences between the flux and bulk values of the transport parameters for the Ar^+ ion swarm in Ar gas.

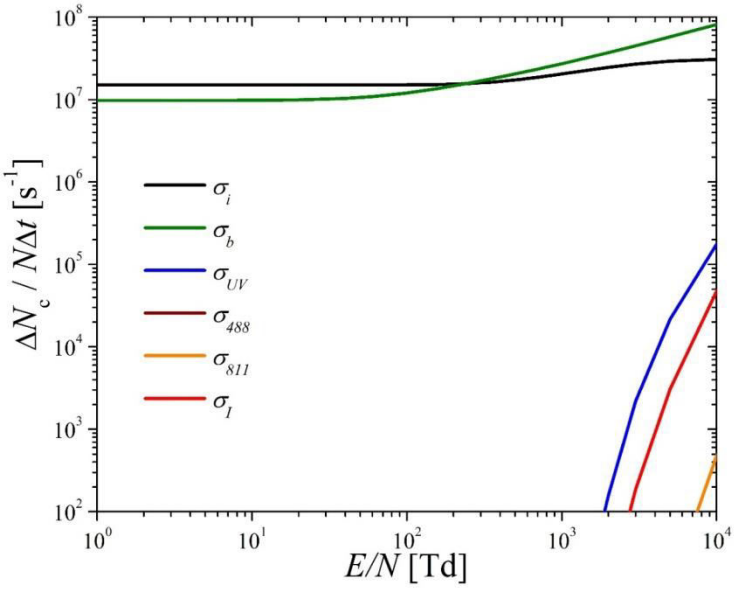


Fig. 2. Rate coefficients for Ar^+ in Ar as a function of E/N at 300 K.

At low E/N values (E/N < 40 Td), the average energy of the swarm is slightly higher than the thermal energy. Fig. 3 presents the ion energy probability function (*IEPF*) for Ar^+ ions in Ar at these low E/N values, which closely follows the Maxwellian energy distribution. The assumption that the ion energy distribution follows a one-dimensional Maxwellian with a

'temperature' T, dependent on E/N, was applied in determining the results from the work of Phelps and Petrović [19].

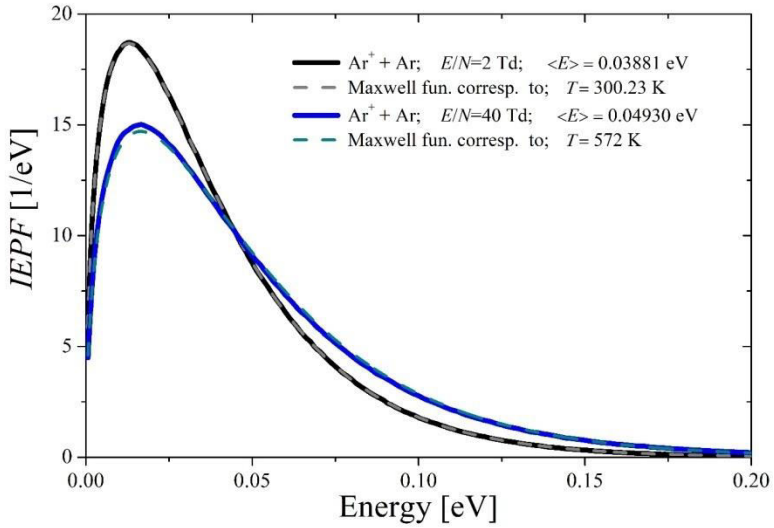


Fig. 3. Ion energy probability function (*IEPF)* for Ar^+ in Ar as a function of low *E/N*.

As *E/N* increases, the *IEDF* deviates from the Maxwellian distribution, as shown in Fig. 4, where the *IEDF* is presented for *E/N* = 100 Td and *E/N* = 500 Td. As *E/N* grows, the impact of backward elastic collisions becomes more pronounced, leading to a significant formation of slow Ar^+ ions. This effect is clearly reflected in the shape of the *IEPF* at *E/N* = 100 Td and is especially evident at *E/N* = 500 Td.

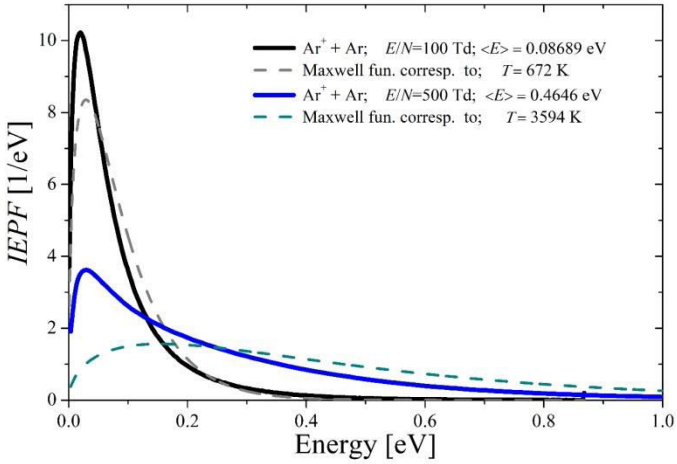


Fig. 4. Ion energy probability function (*IEPF)* for Ar^+ in Ar as a function of moderate *E/N*.

At high values of *E/N*, the *IEDF* exhibits a significant number of ions close to thermal, causing a considerable deviation from the Maxwellian distribution, as shown in Fig. 5 at *E/N* = 10000 Td. In this case, the assumption that the energy distribution follows a one-dimensional Maxwellian with a temperature *T* determined by *E/N* is not valid. Therefore, it is necessary to solve the Boltzmann equation for this system [20] or to apply precise Monte Carlo simulations, as done in this work.

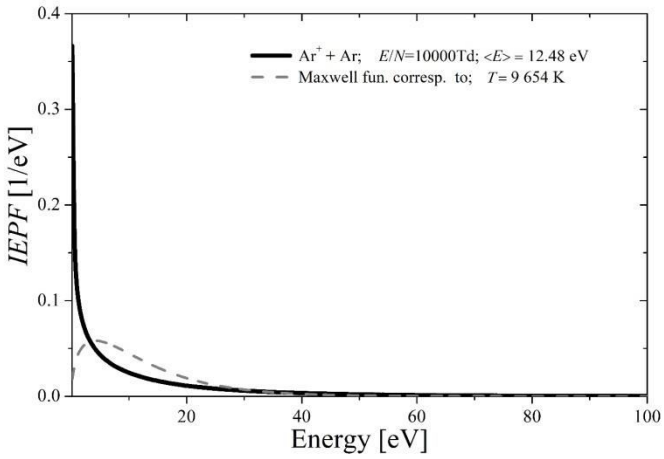


Fig. 5. Ion energy probability function (*IEPF)* for Ar^+ in Ar as a function of high *E/N.*

The bulk and flux drift velocities [21, 22] as a function of *E/N* are shown in Fig. 6. A comparison with both theoretical and experimental results from works [19, 23] is also provided. The drift velocities in this paper, obtained from Monte Carlo simulations in both real space (bulk) and velocity space (flux), were calculated as d*x*/d*t* and <*v*>, respectively. In the absence of non-conservative collisions, such as attachment or ionization, no deviations were observed between the bulk and flux drift velocities. Additionally, the figure demonstrates a good agreement between the drift velocities reported in [19] and the experimental data, where a Maxwellian *IEDF* is assumed at low *E/N*. However, at high *E/N*, significant deviations are evident.

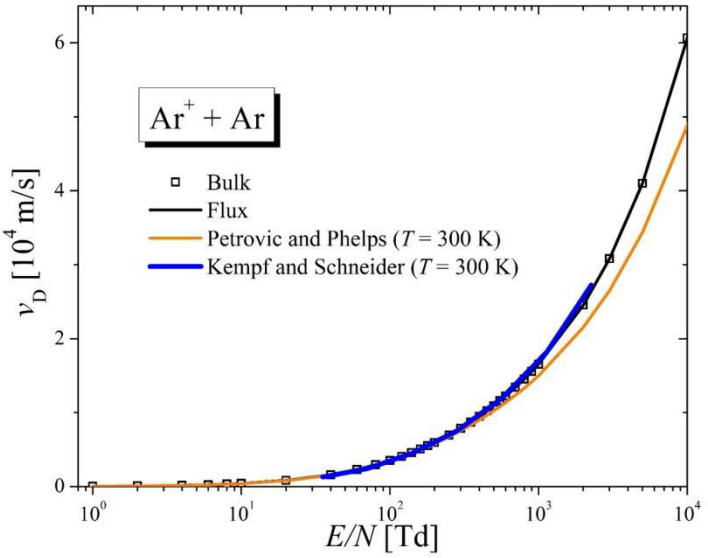


Fig. 6. Bulk and flux drift velocity for Ar^+ in Ar at 300 K.

The reduced mobility of Ar^+ ions as a function of *E/N* is shown in Fig. 7. The presented values [24–27] were obtained using bulk drift velocities, as these are most commonly measured in experiments. However, appropriate quantities should be applied depending on the source of experimental data [28–33]. At low *E/N*, mobility remains approximately constant. This is a consequence of dominant inductive-dipole interactions

between Ar^+ ions and Ar atoms at low energies, which result in a constant collision frequency independent of energy. As *E/N* increases, the total collision frequency rises, increasing the resistance of the Ar gas to Ar^+ ion drift and causing a decrease in reduced mobility.

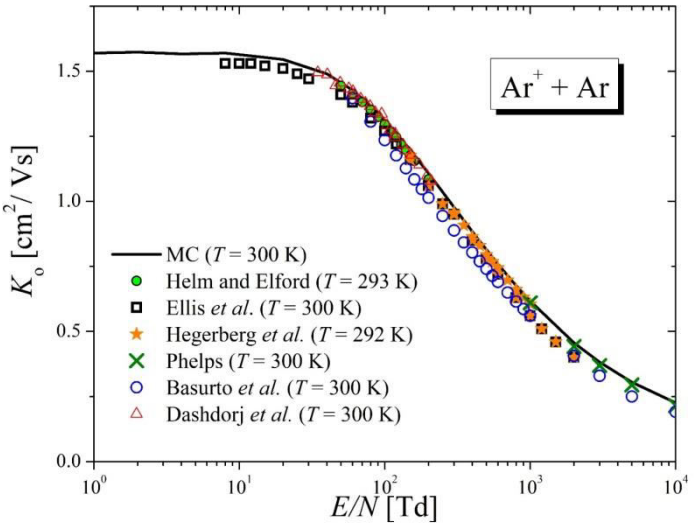


Fig. 7. Reduced mobility for Ar^+ in Ar as a function of *E/N*.

Ion mobility (*K*) is defined as the proportionality factor between the ion drift velocity and the electric field. One often exploits the reduced or standard mobility defined as:

$$K_0 = \frac{v_d}{N_0 E} N \qquad (1)$$

where v_d is the drift velocity of the ion, *N* is the gas density at elevated temperature *T*, $N_0 = 2.69 \cdot 10^{25}$ m^{-3} and *E* is the electric field. Agreement with experimentally determined values is more than satisfactory [34, 35].

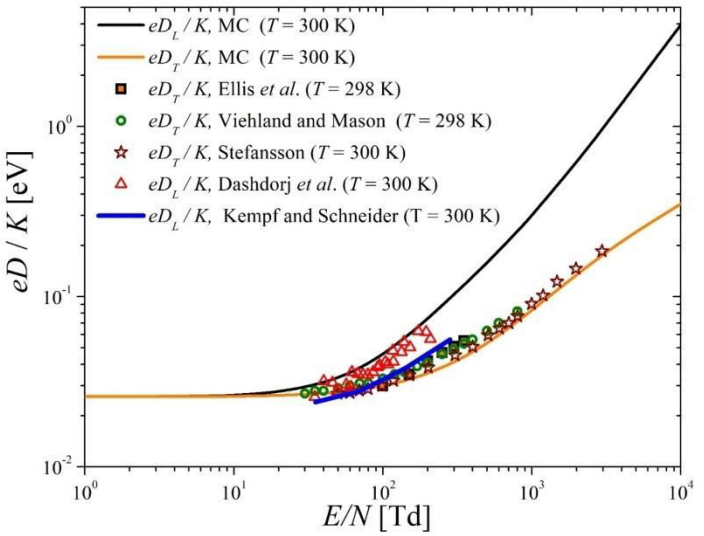


Fig. 8. Characteristic energy for Ar^+ in Ar as a function of *E/N*.

In Fig. 8 we show the characteristic energies (diffusion coefficient normalized to mobility, eD/K, in units of eV) based on transversal (D_T) i longitudinalni diffusion coefficients (D_L). No differences between bulk and flux values are observed. The agreement with the experimental results is good [36, 37].

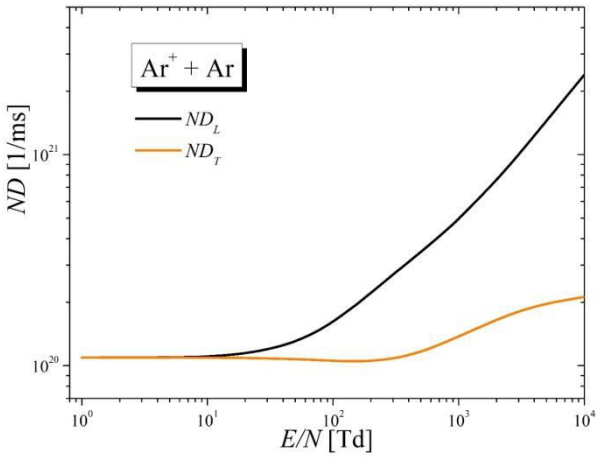


Fig. 9. Longitudinal diffusion coefficients for Ar^+ in Ar as a function of E/N.

The diffusion coefficients ND_L and ND_T are shown in Fig. 9. At low values of E/N, the transverse and longitudinal diffusion coefficients are approximately equal, causing the swarm to expand spherically through space at low electric fields, while its center of mass drifts along the field direction. The spatially resolved average kinetic energy along the electric field shows a linear increase with position, resulting from the potential difference between the front and tail edges of the swarm. In contrast, the spatially resolved average kinetic energy perpendicular to the electric field follows a parabolic shape [38]. These differences in the spatial dependence of the average kinetic energy give rise to the disparity between ND_L and ND_T. When the collision frequency increases with energy, as in the case of the Ar^+ + Ar system, more collisions occur at the front of the swarm compared to its tail, reducing longitudinal diffusion relative to transverse diffusion. As a result, the swarm compresses along the field direction and expands in the form of an ellipsoid while drifting along the field direction.

4. Conclusion

In addition to presenting the data, we analyze the effects of backward elastic collisions on ion transport. These reactions are found to have a significant impact, substantially reducing both ion mobility and the longitudinal diffusion of the swarm. Data for ion swarm parameters [39, 40] are required for hybrid and fluid codes, and the current focus on liquids or liquids in rare gas mixtures dictates the need to produce data compatible with those models.

Acknowledgments

The authors acknowledge funding provided by the Institute of Physics University Belgrade, through the grant by the Ministry of Education, Science and Technological Development of the Republic of Serbia No. 451-03-136/2025-03.

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Сажетак: *У овом раду представљамо комплетан скуп пресека и транспортне коефицијенте Ar^+ јона у Ar гасу. Ови подаци су потребни за моделовање у бројним апликацијама од технолошког значаја. Метода Монте Карло симулације је примењена за прецизно израчунавање транспортних параметара у хидродинамичком режиму. Разматрамо нове податке за Ar^+ јоне у Ar гасу као што су брзина дрифта, карактеристична енергија, флукс и балк редукована мобилност, брзински коефицијенти, коефицијенти дифузије и други транспортни параметри дати као функција ниских и умерено редукованих електричних поља E/N (E-јачина електричног поља, N - концентрација гаса). Даље, показано је да функција вероватноће енергије јона прати Максвелову дистрибуцију при електричним пољима са малим интезитетом електричног поља, док при електричним пољима са великим интезитетом значајно одступа.*

Кључне речи*: Монте Карло Симулације, Ar+ јони, Ar гас.*

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